

7.1)

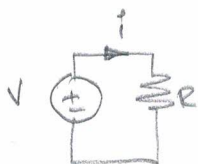
MEDDELEFFREKT $P_{medel} = V_{rms} i_{rms} =$
 $= V_{rms} \frac{V_{rms}}{R} = V_{rms}^2 \frac{1}{R}$

NUMERISKT: $P_{medel} = 90^2 \frac{1}{20} = \underline{405 \text{ W}}$

7.2)

MEDDELEFFREKT $P_{medel} = V_{rms} i_{rms} = V_{rms}^2 \frac{1}{R}$
 $R = \frac{V_{rms}^2}{P_{medel}} = \frac{240^2}{1000} \approx \underline{57.6 \Omega}$

7.3)



$R = 50 \Omega$ EFFEKT?

$P = V_{rms} i_{rms} = \frac{1}{2} V_m i_m$
 MEN $V_m = R i_m \Rightarrow P = \frac{1}{2} R i_m^2$

a) $i_m(t) = 7 \cos(100t) \text{ A}$

$P = \frac{1}{2} 50 \cdot 7^2 \approx \underline{1225 \text{ W}}$

b) $i_m(t) = 7 \cos(100t - 30^\circ) \text{ A}$

$P = \frac{1}{2} 50 \cdot 7^2 \approx \underline{1225 \text{ W}}$

c) $i_m(t) = 7 \cos(100t) - 3 \cos(100t - 60^\circ)$, HOPPA TIL
 KOMPLEX FORM

$I = 7 \angle 0^\circ - 3 \angle -60^\circ = 7 \cos(0^\circ) - 7 \sin(0^\circ) j -$
 $- 3 \cos(-60^\circ) - 3 \sin(-60^\circ) j =$
 $= 5.5 + j 2.598 \approx 6.083 \angle 25.28^\circ$

$\Rightarrow i_m = 6.083 \text{ A}$

$\Rightarrow P = \frac{1}{2} 50 \cdot 6.083^2 = \underline{925.1 \text{ W}}$

d) $i_m(t) = 7 \cos(100t) - 3 \text{ A}$

MOMENTAN SPÄNNING ÖVER R RÖR

$V(t) = R i_m(t) = 350 \cos(100t) - 150 \text{ VOLT}$

MOMENTAN EFFEKT

$P(t) = V(t) i(t) = (350 \cos(100t) - 150) (7 \cos(100t) - 3)$

$= 2450 \cos^2(100t) - 1050 \cos(100t) - 1050 \cos(100t) + 450$

MEDELVÄRDEGIVNA

$P_{medel} = \frac{1}{T} \int_0^T [2450 \cos^2(100t) - 2100 \cos(100t) + 450] dt$

↙

$$P_{\text{mean}} = \frac{1}{T} \int_0^T \left[\frac{2450}{2} (1 + \cos(200t)) - 2150 \cos(100t) + 450 \right] dt =$$

$$= \frac{1}{T} \left(\frac{2450}{2} T + 450 T \right) = \underline{1625 \text{ W}}$$

SVAR = a) $P = 1225 \text{ W}$

b) $P = 1225 \text{ W}$

c) $P \approx 925.1 \text{ W}$

d) $P \approx 1625 \text{ W}$